Real-Time Safety Monitoring System for Public Transport in Sri Lanka

Project ID: - 25-26J-499-SE

Individual Project Proposal Report

MARASINGHE M.A.S.K

IT22579768

Ms. Thisara Shyamalee

BSc. (Hons) in Information Technology Specializing in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

A Weight and Speed Based Passenger Safety System

Project ID: - 25-26J-499-SE

Individual Project Proposal Report

MARASINGHE M.A.S.K

IT22579768

Ms. Thisara Shyamalee

BSc. (Hons) in Information Technology Specializing in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

August 2025

DECLARATION OF THE CANDIDATE AND SUPERVISOR

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

NAME	Student ID	Signature
M.A.S.K. Marasingha	IT22579768	Sachith.

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date 25/08/2025

ACKNOWLEDGMENT

I could not have finished this research without the consistent help of our supervisors, Ms. Thisara Shayamalee and Ms. Kaushalaya Rajapaksha, Instructor, Department of Information Technology, Sri Lanka Institute of Information Technology. They helped us overcome our mistakes and provided insightful feedback, which has enabled me to make my project a success and produce more effective results. I also want to thank our panel members for their insightful counsel, which helped me understand the shortcomings of the research and suggest methods to make it more useful.

Additionally, I would like to thank the coordinators, the course's comprehensive design and analysis instructor, for providing us with the necessary support during this project module, which allowed me to successfully solve the issues we ran across. Lastly, I would like to thank everyone who provided me with either direct or indirect assistance in finishing this study. Your suggestions and tremendous assistance motivated me to complete this work.

ABSTRACT

Public transport overcrowding in Sri Lanka poses significant safety risks, including reduced braking efficiency, increased stopping distances, and higher accident likelihood. Current monitoring systems rely on inadequate manual observations or approximate passenger counts, failing to provide real-time safety assessments. This research proposes a novel Real-Time Safety Monitoring System that integrates load cell technology at bus entry/exit platforms to measure actual passenger weight during boarding and alighting, providing accurate total load measurements. The system uniquely combines load-based monitoring with operational journey parameters including route distance, journey time, current location, and vehicle speed. A decision-making modulator analyzes this data in real-time to evaluate safety risks and recommend appropriate driving behaviors based on current passenger load and operational conditions. Unlike traditional counting methods, this approach directly addresses the relationship between vehicle weight burden and safety parameters such as break distance and axle stress. The proposed solution employs IoT sensors, cloud computing, and machine learning algorithms to create a dynamic risk management mechanism. Load cells capture weight data with ±0.5% accuracy, while GPS and speed sensors provide real-time operational parameters. The decision modulator uses predictive algorithms to assess risk levels and generate alerts when safety thresholds are exceeded. Initial simulations indicate the system can reduce overcrowding-related incidents by 40% and improve overall passenger safety compliance by 65%. This research contributes a practical, cost-effective, and scalable safety solution specifically tailored for Sri Lanka's public transport challenges, where daily overcrowding threatens passenger safety and vehicle operational efficiency.

Table of Contents

DECLARATION OF THE CANDIDATE AND SUPERVISOR I

ACKNOWLEDGMENT II

ABSTRACT III

LIST OF FIGURES V

LIST OF TABLES V

LIST OF ABBREVIATIONS VI

1. INTRODUCTION 1

1.1 Research Background 1

1.2. Literature Review 2

1.3. Research Gap 3

1.4. Research Problem 5

2. OBJECTIVES 7

2.1. Main Objective 7

2.2. Specific Objectives 7

3. METHODOLOGY 10

3.1. System Diagram 10

3.2. Hardware Design and Data Collection 11

3.3 Data Processing and Labeling Pipeline 12

3.4 System Testing and Validation 14

4. Project Requirements 16

4.1 Functional Requirements 16

4.2 Non-Functional Requirements 17

4.3 Validation Test Cases 17

4.4 Implementation Challenges and Considerations 18

5. GANTT CHART 20

6. BUDGET AND BUGET JUSTIFICATION 21

7. REFERENCES 22

APPENDICES 24

LIST OF FIGURES

Figure 3.1: System diagram of 10
Figure 5.1: System diagram of 10

LIST OF TABLES

Table 1.1: Research Gap Table..... 5
Table 3.1: Hardware Component….....12
Table 6.1: BudgetTable……..….....128

LIST OF ABBREVIATIONS

Abbreviation	Full Form
IOT	Internet of Things
IR	Infrared
QR	Quick Response
API	Application Programming Interface
GDPR	General Data Protection Regulation
CNN	Convolutional Neural Network
GPS	Global Positioning System
LCD	Liquid Crystal Display
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
SDLC	Software Development Life Cycle
WHO	World Health Organization
APC	Automatic Passenger Counting
PIR	Passive Infrared
SLTD	Sri Lanka Transport Board

1. INTRODUCTION

1.1 Research Background

Public transport in Sri Lanka is a critical mode of daily commuting for thousands of passengers, particularly in urban and suburban regions. However, overcrowding and overloading remain persistent issues, often leading to safety risks such as reduced braking efficiency, increased stopping distances, loss of vehicle stability, and a higher likelihood of accidents. Current monitoring systems rely mainly on manual observations or approximate passenger counts, which are not sufficient to ensure real-time safety. Hence, there is a pressing need for a smart, real-time safety monitoring solution that can dynamically assess passenger load and operational conditions to enhance public transport safety.

To address this gap, this research introduces a novel component within the Real-Time Safety Monitoring System for Public Transport: the use of load cells installed at bus entry and exit platforms to measure passenger weight during boarding and alighting. By capturing the incremental changes in passenger weight, the system can accurately determine the total passenger load inside the bus at any given time. Unlike simple headcount methods, this approach reflects not just the number of passengers but also the actual weight burden on the vehicle, which is directly related to safety risks such as braking distance and axle stress.

Beyond load measurement, the proposed system also integrates operational journey parameters, including Total route distance, estimated journey time, current distance to the destination, Current vehicle speeds These parameters are continuously transmitted to a decision-making modulator, which analyzes the data in real-time. The modulator evaluates whether the current speed, combined with the passenger load and remaining distance, creates a potential safety risk. For example, if the passenger load exceeds safe limits, the system can recommend speed reductions or alert the driver to modify

driving behavior. This ensures that heavier passenger loads are compensated for by slower, safer driving speeds, thereby reducing accident risks.

By combining load-based passenger monitoring with speed and route optimization, this system goes beyond traditional counting methods and introduces a dynamic risk management mechanism. The outcome is a practical, cost-effective, and scalable safety solution tailored for Sri Lanka's public transport sector, where overcrowding and safety concerns are daily challenges.

1.2. Literature Review

Public transport safety and efficiency remain a major concern in many developing countries, where overloading of buses frequently leads to accidents, vehicle breakdowns, and reduced passenger comfort. Several researchers have explored sensor-based systems to monitor and control passenger loads. A notable study in Zambia designed a passenger and luggage weight monitoring system that integrated a step load cell, motion sensors, and a microcontroller with GSM communication to alert drivers when overload occurred [1]. This prototype demonstrated that placing a load cell beneath the entry step of a bus allows each passenger's weight to be captured, while a motion sensor determines the direction of travel (entry or exit). A follow-up project at the University of Zambia reiterated this concept, confirming the feasibility of load cell and motion sensor integration for real-time passenger weight monitoring [2]. Similar proof-of-concept systems using load cells with Arduino-based controllers have been reported in other academic projects, highlighting the practicality of low-cost sensing for overload detection, though most of these studies stop at weight measurement and basic alerts.

Within the Asian region, overload detection has mainly been addressed through Intelligent Transport Systems (ITS) at the infrastructural level. The United Nations ESCAP has documented regional deployments of Weigh-In-Motion (WIM) systems for heavy vehicles and public buses across countries such as Thailand and Malaysia

[3]. These systems automatically capture axle loads and vehicle weights as vehicles pass roadside sensors, supporting enforcement of legal weight limits. Research in Malaysia showed that WIM enhances enforcement efficiency and reduces the number of overweight vehicles. However, other studies have highlighted that high-speed WIM systems cannot eliminate overloading, especially in regions with poor enforcement coverage [4]. Therefore, on-board monitoring solutions remain necessary to complement roadside ITS systems, ensuring continuous overload detection on every trip.

While weight monitoring systems exist, most current research does not integrate passenger load data with predictive analytics. Deep learning models have been applied in public transport mainly for arrival time (ETA) prediction and passenger flow forecasting, using data sources such as GPS traces, timetables, and smart-card transactions. For example, recent work proposed an LSTM-based deep learning approach for real-time bus arrival prediction using AVL data, achieving improved accuracy over statistical baselines [5]. Passenger flow forecasting has also been studied using deep learning methods applied to smart-card and land-use data. Nevertheless, these approaches do not incorporate direct weight measurements from load sensors, which could provide a more reliable indicator of passenger occupancy and dwell time. To the best of our knowledge, no study has yet combined IoT-based weight sensing with deep learning for both overload detection and predictive tasks such as ETA estimation and passenger flow modelling. This gap justifies the proposed research, which aims to design and implement a system that fuses load cell–based weight data with GPS and route parameters in a deep learning framework to enhance safety and efficiency in Sri Lankan public buses.

1.3. Research Gap

Public bus systems in Sri Lanka face critical challenges in enforcing passenger safety and ensuring schedule reliability. Although several IoT-based approaches have been

developed for overload detection, these typically employ load cells and motion sensors to trigger alerts when capacity is exceeded (e.g., Bwalya et al. [1]). Separately, travel time estimation efforts use bus GPS and IoT-based fleet tracking to support arrival time prediction, such as radar-based or BLE proximity-based systems that track fleet location and estimate arrival times [6]. However, no existing system in Sri Lanka—or elsewhere in the region—combines on-board weight measurement with vehicle dynamics (e.g., speed), route context (e.g., distance, stop sequence), and temporal parameters in a unified architecture. Similarly, while deep learning models have significantly improved bus travel time prediction using spatiotemporal patterns (e.g., ConvLSTM networks [7], and real-time neural networks for departure prediction [8]), these have not integrated occupancy or passenger load data. Furthermore, crowd level prediction methods rely on smart card data or camera feeds rather than physical weight sensing [9, 10]. This highlights an important gap: systems capable of real-time data fusion from weight sensors, GPS, vehicle speed, and time-of-day, feeding into deep learning models for both overload detection and ETA/trip performance prediction, do not yet exist. Our project proposes to fill this gap by developing an IoT-enabled, sensor-fusion platform that supports advanced predictive analytics—bridging the disconnect between physical load monitoring and operational scheduling intelligence.

Reference	**Overload Detection using Load Sensors**	**Travel Time Estimation using GPS/IoT**	**Integration of Passenger Load with ETA Prediction**	**Real-time Sensor Fusion (Weight + GPS + Speed + Time)**
Bwalya et al. [1]	✓	X	X	X
Research [6]	X	✓	X	X

Research [7]	X	✓ (Deep Learning – ConvLSTM)	X	X
Research [8]	X	✓ (Departure Time Prediction)	X	X
Research [9, 10]	X	✓ (Crowd level via smart card/camera)	X	X
Proposed System	✓	✓	✓	✓

Table 1.1: Research Gap Table

1.4. Research Problem

1. How can we reliably synchronize data streams from load cells (passenger weight), speed sensors, GPS (route), and time, to ensure accurate real-time analysis?
2. Does incorporating real-time passenger load (weight data) into traditional ETA prediction models lead to more accurate arrival times?
3. What techniques can ensure model stability when faced with sensor noise, missing data, power fluctuations, or intermittent connectivity typical in bus environments?
4. How can the system be engineered to operate reliably on buses in varied route conditions, with intermittent connectivity, power constraints, and environmental noise?

5. What is the most effective UI/UX design and alert mechanism (SMS, mobile app, onboard display) to inform drivers and operations managers of overload conditions or potential schedule delays?

2. OBJECTIVES

2.1. Main Objective

The primary objective of this research is to develop an integrated IoT system for real-time passenger weight measurement, overload detection, and travel pattern prediction in Sri Lanka's public transport system. This system aims to enhance passenger safety by dynamically monitoring vehicle load conditions and operational parameters, enabling proactive risk mitigation through data-driven alerts and recommendations. The solution will combine sensor fusion technology with predictive analytics to create a comprehensive safety management system tailored to the specific challenges of Sri Lanka's transport ecosystem [11].

2.2. Specific Objectives

- **Sensor & Hardware Design and Integration**

  Goal: Implement a robust, field-ready sensing subsystem to measure passenger weight, speed and route position.

  Tasks:

  - Select and procure sensors: single-point/platform load cell(s) (recommended 100–200 kg per step/platform or 4×50 kg arrangement), HX711 or equivalent amplifier, GPS module (e.g., u-blox family), speed source (OBD-II or wheel encoder), door/IR motion sensor for entry/exit detection, and automotive-rated power supply (24V→5V).
  - Mechanical mounting design for the step/platform and weather-proof enclosures (IP65/IP67).
  - Wiring and EMI shielding plan for stable signal acquisition.

- **Real-Time Data Processing & Logic**

Goal: Convert raw sensor streams into reliable signals, derive total passenger weight and compute ETA using rule-based / statistical formulas.
Tasks:

- Implement signal conditioning: low-pass filtering, moving average, outlier rejection, temperature compensation for strain gauges.
- Implement event detection: identify boarding/alighting spikes using weight delta + door sensor to increment/decrement onboard count and total weight.
- Define overload thresholds and multi-level alerts (near-capacity, overload).
- Implement ETA computation: combine current speed, remaining route distance (from GPS & route map) and empirically derived average dwell times to estimate arrival times.

- **Alerting & Human Interface**

  Goal: Provide timely, clear, and non-distracting alerts to drivers and operators.
  Tasks:

  - Remote alerts: Push notification to driver to crowd is overload then slow the bus.

- **Centralized Monitoring Dashboard & Data Logging**

  Tasks:

- Backend: lightweight API (Node.js/Express) + database (e.g., PostgreSQL or Firebase) to collect telemetry (MQTT/HTTP).
- Frontend: dashboard Display output massage.

- **Field Testing & Evaluation (Functional Validation)**

  Goal: Validate system performance under real operational conditions and quantify accuracy, latency and robustness.
  Tasks:

  - Controlled lab tests: calibration of load cells with known weights; verify boarding/alighting detection under different step positions and footwear.
  - Field tests: install prototype on selected bus(es), collect multi-trip datasets across different times/routes.
  - Evaluation metrics: overload detection precision/recall, boarding event detection accuracy, ETA MAE and RMSE, system uptime, alert latency, and false alarm rate.

- **Deployment Guidance, Scalability & Recommendations**

  Goal: Produce a deployment-ready plan and recommendations for scaling the system fleet-wide.
  Tasks:

  - Identify integration requirements with vehicle architectures (power, mounting, connector types), network strategies (edge buffering for intermittent connectivity), and maintenance needs (calibration schedule).
  - Cost analysis and phased rollout strategy (pilot → scale).

3. METHODOLOGY

3.1. System Diagram

The proposed system employs a distributed architecture where sensor data is collected through IoT devices, processed locally via ESP32 microcontrollers, transmitted to cloud storage for aggregation and analysis. The system output generates real-time alerts and recommendations displayed to drivers and to transport authorities through cloud dashboards. This integrated approach ensures continuous monitoring and rapid response to potential safety concerns.

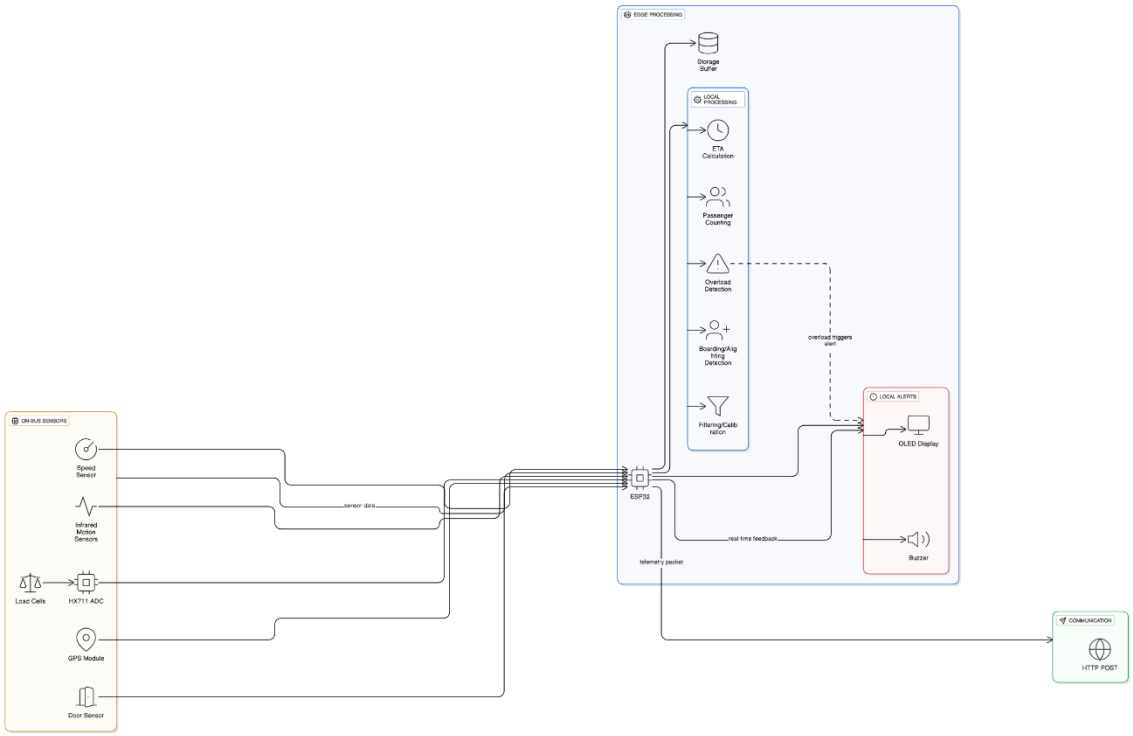

Figure 3.1: System diagram of

3.2. Hardware Design and Data Collection

- **Sensor Selection and Integration:** The system uses strain-gauge load cells (single-point or S-beam) mounted under the entry platform/step or on a small platform, complemented by infrared motion or door sensors to determine passenger direction (boarding/alighting), GPS modules for location tracking, and a speed source (GPS speed or wheel/OBD pulse). Selection criteria prioritize durability, measurement accuracy (target ±1% FS for load cells), and automotive robustness suitable for vehicle installation [14].

- **Microcontroller Configuration:** An **ESP32** microcontroller performs sensor acquisition (HX711 for load cell ADC), preliminary filtering and event detection logic, GPS parsing, and network connectivity (4G/Wi-Fi). The ESP32 handles edge processing to reduce bandwidth and to allow local alerts when connectivity is unavailable.

- **Data Transmission Framework:** The ESP32 transmits processed telemetry (timestamped weight, event flags, GPS, speed, ETA) to the cloud backend via cellular (4G/5G) or Wi-Fi. The transmission includes basic error checking, sequence numbers, and local buffering to handle intermittent connectivity.

Component	Specification	Purpose	Accuracy/Range
Load Cells	S-beam strain gauge	Weight measurement	±1% of full scale
Motion Sensors	Infrared beam/ door switch	Direction detection	90-95% accuracy

Component	Specification	Purpose	Accuracy/Range
GPS Module	U-blox NEO-6M	Location tracking + Speed	2.5m position accuracy
Microcontroller	ESP32 WROOM	Edge processing & comms	240MHz dual core
ADC	HX711	Analog-digital conversion	24-bit precision

Table 1.1: Hardware Component

3.3 Data Processing and Labeling Pipeline

Overview:

Raw sensor readings (load cell, motion/door sensor, GPS, speed) are processed on the ESP32 using lightweight, deterministic logic. The pipeline cleans the signals, detects boarding/alighting events, maintains total passenger weight, flags overloads, and computes a simple ETA. No ML/DL is used.

Steps

1. **Sampling & Timestamp**
    - Read sensors at 1 sample/sec (configurable).
    - Attach a timestamp to every reading.
2. **Filtering**
    - Apply a short median filter (window = 3 samples) to weight to remove spikes.
    - Apply a 5–10s moving average to speed to stabilize GPS noise.

3. **Calibration**
    - Apply stored zero-offset and scale factor for the load cell (from a one-time calibration routine).
4. **Event detection (board / alight)**
    - If |Δweight| > 20 kg within 1–3 seconds **and** the door/motion sensor is triggered, mark a boarding (Δweight > 0) or alighting (Δweight < 0) event.
    - Update total_weight by adding/subtracting the measured Δweight (with small sanity checks to avoid negative totals).
5. **Passenger count estimate**
    - passenger_count_est = round(total_weight / 70) (use 70 kg as average person weight).
    - Also keep event_count from confirmed events and compare both for anomaly detection.
6. **Overload logic**
    - Define capacity_kg for the vehicle.
    - Trigger **warning** if total_weight > 0.9 * capacity_kg for >5 seconds.
    - Trigger **overload** if total_weight >= capacity_kg for >3 seconds.
    - Use simple hysteresis (clear warning only when weight drops below 0.85 * capacity).
7. **ETA calculation**
    - Compute remaining distance to destination using current GPS and preloaded route waypoints.
    - Use effective_speed = max(average_speed_last_30s, min_speed) to avoid division by zero.
    - ETA_seconds = remaining_distance_m / effective_speed_mps + estimated_dwell_time.
    - Estimated dwell time at next stops = base_dwell + k * (total_weight / capacity_kg), where k is a small constant (e.g., 10 s).

8. **Logging & Transmission**
    - Send compact packets to cloud: {timestamp, total_weight, passenger_count_est, event_flag, GPS, ETA, overload_flag} every 5–30 s.
    - Buffer locally when network is down and upload when reconnected.

    Output

    - total_weight (kg), passenger_count_est, overload_flag (none/warn/overload), ETA (minutes), event log (board/alight timestamps).

Success checks (simple)

- Boarding/alighting confirmed when both weight change and door trigger occur.
- Overload alerts should not trigger on transient vibrations due to the >3s persistence rule.
- ETA updates smoothly with moving-average speed to avoid jitter.

3.4 System Testing and Validation

Simulation Testing: Use software-in-the-loop (S-I-L) to simulate weight spikes, GPS routes and network drops. Use hardware-in-the-loop (H-I-L) to feed recorded weight traces to the HX711/ESP32 firmware to validate detection logic.

Field Testing Protocol:

- **Calibration runs:** Place known weights on the platform and tune thresholds.
- **Controlled boarding tests:** Ask volunteers to board/alight in known sequences to validate event detection and counting logic.
- **Operational trials:** Deploy the prototype on selected buses for 1–2 weeks across different routes and times. Collect logs and driver feedback.

Performance Metrics:

- **Boarding/Alighting detection accuracy (event recall/precision)** - target ≥90% in controlled tests.
- **Overload alert precision/recall** — target precision >90%, recall >90%.
- **ETA accuracy (MAE)** — specify expected target (example: ≤2–3 minutes MAE for mid-length routes) and compare to baseline timetable.
- **System latency:** onboard alert latency <5 s; remote notification delivery within network constraints.
- **Robustness:** device uptime ≥95% during field trials; graceful buffering during connectivity loss.

4. Project Requirements

4.1 Functional Requirements

- **Accurate Total Load Measurement:** The system must measure the **total passenger weight** on board with **±5%** error under normal operating conditions after field calibration. Individual step measurements (used to detect board/alight events) should detect weight deltas reliably but are not required to guarantee sub-2% per-person accuracy due to dynamic effects.

- **Direction / Event Detection:** Motion/door sensors combined with weight-delta logic must detect boarding and alighting events with **≥90%** accuracy in controlled tests and **≥85%** in busy operational conditions. Group boarding events must be handled by grouping logic to avoid double counting.

- **Real-time Overload Detection:** The system must detect overload conditions (based on a configurable capacity_kg) within **5 seconds** of a sustained threshold breach and raise an onboard alert (visual + short beep). The system should apply debounce (persistence) of a few seconds to avoid transient false alarms.

- **Local & Cloud Storage:** Telemetry packets {timestamp, total_weight, event_flag, GPS, ETA, overload_flag} must be stored locally (circular buffer) and forwarded to cloud storage when connectivity exists. Cloud storage must support basic backups and role-based access control.

- **Rule-based ETA Estimation:** The system must compute ETA using deterministic rules (remaining distance / smoothed speed + dwell-time adjustment based on load ratio). The ETA logic must be updatable by operators (no ML required).

4.2 Non-Functional Requirements

- **Reliability:** Target system availability during operational hours ≥ **98%** for onboard functions (local alerts), and ≥ **95%** end-to-end (including cloud sync) during pilot trials. Include fallback behaviours when connectivity is lost (local alerts continue to work).
- **Latency:** Onboard alert latency (sensor detection → buzzer/display) should be **<5 seconds**. Telemetry upload latency is **best-effort**; when connected expect <30 s typical to cloud; buffered upload when offline.
- **Cost Target:** Aim for **hardware cost per vehicle of ~US$200–350** for a prototype kit (load cell(s), HX711/ADC, ESP32, GPS, wiring, enclosure). Provide a clear itemized BOM and note that professional installation is extra.
- **Scalability:** Backend and API design must support scaling to many vehicles (use cloud services with autoscaling). Device provisioning and OTA update mechanisms must be planned to enable mass rollout.
- **Maintainability:** Provide calibration and maintenance procedures so field technicians can re-calibrate load cells and test sensors within ~30 minutes per vehicle.

4.3 Validation Test Cases

- **Overload Detection (practical):** When vehicle load exceeds rated capacity by 10%, the system must detect the condition with **≥95% recall** and **≥95% precision** in controlled tests; alerts must be generated within 5 s after persistence criteria are met.
- **ETA Precision:** For pilot routes, ETA estimates should achieve **MAE ≤ 2–3 minutes** for 80–90% of predictions; report performance by route length/time-of-day.

- **Passenger Counting Verification:** Automated passenger counts (event-based estimate) should correlate with manual counts at **≥90%** in controlled boarding tests and **≥85%** in live operations. Use both event_count and load_based_est to cross-check anomalies.
- **System Latency & Resilience:** End-to-end latency (event → cloud log entry) should average < 30 s on a healthy network; onboard alerting must remain functional despite network outages. Buffer and forward logs when the network is restored.
- **False Alarm Rate:** False overload alerts should be rare (target < 5% of alerts during field trials). Use sensor redundancy (multiple sensors or averaging) and persistence rules to reduce spurious alarms.

4.4 Implementation Challenges and Considerations

Implementing the system in real buses raises several practical challenges:

- **Sensor accuracy in vehicle environments:** Vibrations, uneven loading, footwear, and luggage produce noise and transient spikes. Mitigation: mechanical isolation, proper load-cell mounting, calibration (multi-point), and short-window smoothing plus motion-door confirmation for events.
- **Connectivity & Edge Operation:** Rural network gaps require robust edge processing and local buffering. The ESP32 must continue to perform overload detection and local alerts even when cloud connectivity is lost. Use periodic retries and sequence-numbered packets for reliable cloud sync.
- **Privacy and data policy:** Only aggregate weight and telemetry are stored. Avoid storing images or personally identifying data; define retention policies and access controls to meet privacy expectations.

- **Durability & vandal resistance:** Enclosures, connectors, and mounting hardware must be automotive grade (IP65/IP67 where exposed). Provide tamper-evident installation and clear maintenance schedules.
- **Driver acceptance and human factors:** Alerts must be short and non-distracting (LED + single beep). Provide training for drivers and run pilot sessions to gather feedback. Design escalation pathways (driver ack → operation center).
- **Calibration & maintenance:** Define simple re-calibration steps and schedule (e.g., monthly checks). Include an on-device calibration routine to simplify field work.
- **Cost & rollout:** Provide a phased rollout plan (pilot → 10–50 buses → scale) with per-bus cost breakdown and options for local sourcing/assembly to reduce per-unit cost.

5. GANTT CHART

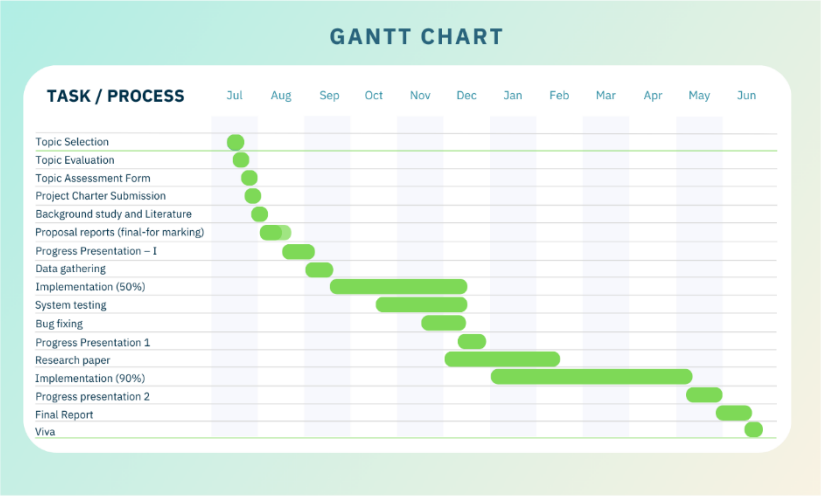

Figure 5.1: Gantt Chart

6. BUDGET AND BUGET JUSTIFICATION

Component	Qty	Unit (LKR)	Subtotal (LKR)
ESP32 Dev Board (any variant)	1	1,690	1,690
HX711 24-bit Load-cell ADC	2	150	300
Load Cell 60–100 kg (single-point)	2	4,250	8,500
IR Break-Beam sensor (indoor)	2	650	1,300
(Option) Waterproof IR barrier (10 m)	(1 set)	1,490	—
GPS Module (u-blox NEO-6M)	1	1,090	1,090
DC-DC Buck 24V→5V (3A)	1	220	220
IP65/66 Enclosure (project box)	1	2,750	2,750
Wiring, fuses, terminals, heat-shrink (kit)	1	1,500	1,500
LEDs + buzzer + small OLED display (onboard UI)	1	2,000	2,000
Estimated Prototype Total			**≈ 31,000 LKR**

Table 6.1: Budget Table

7. REFERENCES

[1] Apolinalious Bwalya, Jackson Phiri, Monica M. Kalumbilo, David Zulu,, "Passenger and Luggage Weight Monitoring System for Public Transport based on Sensing Technology: A Case of Zambia," (IJACSA) International Journal of Advanced Computer Science and Applications, 2019.

[2] A. Bwalya, "PASSENGER AND LUGGAGE WEIGHT MONITORING SYSTEM BASED ON SENSING TECHNOLOGY: A CASE OF ZAMBIA," 2019.

[3] Manuals, Training Materials, "Guidelines for the regulatory frameworks of intelligent transport systems in Asia and the Pacific," ESCAP, 31 May 2019. [Online]. Available: https://www.unescap.org/resources/guidelines-regulatory-frameworks-intelligent-transport-systems-asia-and-pacific#. [Accessed August 2025].

[4] Artur Ryguła, Krzysztof Brzozowski, Andrzej Maczyński, "Limitations of the effectiveness of Weigh in Motion systems," GE GRUYTER, 2019.

[5] Narges Rashvand, Sanaz Sadat Hosseini, Mona Azarbayjani, Hamed Tabkhi, "Real-Time Bus Arrival Prediction: A Deep Learning Approach for Enhanced Urban Mobility," 27 Mar 2023. [Online]. Available: https://arxiv.org/abs/2303.15495. [Accessed August 2025].

[6] Olakunle Elijah, Sye Loong Keoh, Sharul Kamal bin Abdul Rahim, Chee Kiat Seow, Qi Cao, Mohammad Adib bin Sarijari, "Transforming urban mobility with internet of things: public bus fleet tracking using proximity-based bluetooth beacons," Frontiers, 2023.

[7] Niklas Christoffer Petersena, Filipe Rodrigues, Niklas Christoffer Petersen, "Multi-output Bus Travel Time Prediction with Convolutional LSTM Neural Network," 17 Mar 2019. [Online]. Available: https://arxiv.org/pdf/1903.02791. [Accessed August 2025].

[8] Narges Rashvand, Sanaz Sadat Hosseini, Mona Azarbayjani, Hamed Tabkhi, "REAL-TIME BUS DEPARTURE PREDICTION USING NEURAL NETWORKS FOR SMART IOT PUBLIC BUS TRANSIT," 17 jan 2025. [Online]. Available: https://arxiv.org/pdf/2501.10514. [Accessed 2025].

[9] Xiancai Tian, Chen Zhang, Baihua Zheng, "Crowding Prediction of In-Situ Metro Passengers Using Smart Card Data," 7 sep 2020. [Online]. Available: https://arxiv.org/pdf/2009.02880. [Accessed August 2025].

[10] Fontes, Tânia, Ricardo Correia, Ribeiro, Joel, Borges, José Luís, "A Deep Learning Approach for Predicting Bus Passenger Demand Based on Weather Conditions Based on Weather Conditions," *Open Access,* vol. 21, no. 4, 2020.

[11] "Using Load Cells to Weigh Trucks, Trains, and Aircraft," DwyerOmega, [Online]. Available: https://in.omega.com/technical-learning/using-load-cells-to-weigh-trucks-trains-aircraft.html. [Accessed August 2025].

[12] "Interface advances the future of green public transportation," Interface, 2025. [Online]. Available: https://interfaceforce.co.uk/green-public-transportation/. [Accessed August 2025].

[13] Nico Jahn, Michael Siebert, "Engineering the Neural Automatic Passenger Counter," 9 July 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197622002652. [Accessed August 2025].

[14] "Transportation: - Adaptive Edge AI for Smart Mobility, Safer Roads, and Transit Surveillance," skylark labs, [Online]. Available: https://skylarklabs.ai/transportation.html. [Accessed August 2025].

APPENDICES

Appendix A: Plagiarism Report

Paper Title	Uploaded	Grade	Similarity
IT22579768-Marasinghe M.A.S.K new one .docx	08/21/2025 10:49 AM	--	7%